

B5 General Relativity — Model Solutions, Trinity 2023

Paper A16729W1

The rubric instructs the candidate to answer two questions out of four. Full mark-scheme solutions to all four questions are given below.

Question 1 — Weak-field gravity, gravitomagnetism, Nordström's theory

Problem. Linearised Einstein gravity, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, with weak-field Ricci tensor as quoted. Show: (a) static non-relativistic source recovers $\nabla^2\Phi = 4\pi G\rho$ with $h_{00} = -2\Phi/c^2$; (b) the test-particle equation of motion is $\ddot{\mathbf{x}} = -\nabla\Phi + \mathbf{v} \times \mathbf{B}_G$ with $\mathbf{B}_G = \nabla \times c\mathbf{h}$; (c) compare $|\mathbf{B}_G|$ -acceleration to Newtonian gravity at geosynchronous orbit; (d) for Nordström's theory $g_{\mu\nu} = e^{2\Phi/c^2}\eta_{\mu\nu}$ with $R = -\kappa_N g_{\mu\nu} T^{\mu\nu}$, recover $\nabla^2\Phi = 4\pi G\rho$ and find κ_N/κ_E ; (e) explain why light is undeflected and $\mathbf{B}_G = 0$.

(a) Newtonian limit

Adopt signature $(-, +, +, +)$ and write the weak-field Ricci:

$$R_{\mu\nu} = \frac{1}{2}\eta^{\alpha\gamma}(-\partial_\mu\partial_\alpha h_{\nu\gamma} + \partial_\gamma\partial_\alpha h_{\nu\mu} + \partial_\mu\partial_\nu h_{\alpha\gamma} - \partial_\nu\partial_\gamma h_{\alpha\mu}).$$

Specialise to $\mu = \nu = 0$ with $\partial_0 h_{\mu\nu} = 0$:

$$R_{00} = \frac{1}{2}\eta^{\alpha\gamma}(-\partial_0\partial_\alpha h_{0\gamma} + \partial_\gamma\partial_\alpha h_{00} + \partial_0\partial_0 h_{\alpha\gamma} - \partial_0\partial_\gamma h_{\alpha 0}).$$

Drop ∂_0 -terms:

$$R_{00} = \frac{1}{2}\eta^{\alpha\gamma}\partial_\gamma\partial_\alpha h_{00}.$$

Restrict to spatial sum $\alpha = \gamma = i$ (only spatial derivatives survive):

$$R_{00} = \frac{1}{2}\nabla^2 h_{00}.$$

Trace-reverse the field equation by contracting with $g^{\mu\nu}$:

$$R - \frac{1}{2} \cdot 4 R = -\kappa_E T.$$

Solve for R :

$$R = \kappa_E T.$$

Substitute back into the field equation:

$$R_{\mu\nu} = -\kappa_E(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}).$$

Compute the trace of $T_{\mu\nu}$ for the non-relativistic source:

$$T = g^{\mu\nu}T_{\mu\nu} = \eta^{00}T_{00} = (-1)(\rho c^2) = -\rho c^2.$$

Form the source combination:

$$T_{00} - \frac{1}{2}T\eta_{00} = \rho c^2 - \frac{1}{2}(-\rho c^2)(-1).$$

Simplify:

$$T_{00} - \frac{1}{2}T\eta_{00} = \rho c^2 - \frac{1}{2}\rho c^2 = \frac{1}{2}\rho c^2.$$

Insert into the 00-equation $R_{00} = -\kappa_E(T_{00} - \frac{1}{2}Tg_{00})$:

$$\frac{1}{2}\nabla^2 h_{00} = -\frac{1}{2}\kappa_E \rho c^2.$$

Substitute $h_{00} = -2\Phi/c^2$:

$$\frac{1}{2}\nabla^2\left(-\frac{2\Phi}{c^2}\right) = -\frac{1}{2}\kappa_E \rho c^2.$$

Substitute $\kappa_E = 8\pi G/c^4$:

$$-\frac{1}{c^2}\nabla^2\Phi = -\frac{4\pi G}{c^4}\rho c^2.$$

Cancel the c^2 :

$$-\frac{1}{c^2}\nabla^2\Phi = -\frac{4\pi G\rho}{c^2}.$$

Multiply by $-c^2$:

$$\boxed{\nabla^2\Phi = 4\pi G\rho.} \tag{1}$$

(b) Gravitomagnetic equation of motion

Write the geodesic equation with $u^\alpha = dx^\alpha/d\tau$:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu u^\alpha u^\beta = 0.$$

Slow motion: $u^0 \approx c$, $u^i \approx v^i$, $d\tau \approx dt$:

$$\frac{d^2x^i}{dt^2} = -\Gamma_{\alpha\beta}^i \frac{u^\alpha u^\beta}{(u^0)^2} c^2.$$

Keep only terms up to first order in v^i :

$$\frac{d^2x^i}{dt^2} = -c^2\Gamma_{00}^i - 2c\Gamma_{0j}^i v^j.$$

Compute the linearised Christoffel Γ_{00}^i with $\partial_0 h_{\mu\nu} = 0$:

$$\Gamma_{00}^i = \frac{1}{2}\eta^{ii}(\partial_0 h_{i0} + \partial_0 h_{0i} - \partial_i h_{00}) = -\frac{1}{2}\partial_i h_{00}.$$

Compute Γ_{0j}^i similarly:

$$\Gamma_{0j}^i = \frac{1}{2}\eta^{ii}(\partial_0 h_{ij} + \partial_j h_{0i} - \partial_i h_{0j}) = \frac{1}{2}(\partial_j h_{0i} - \partial_i h_{0j}).$$

Evaluate the static piece using $h_{00} = -2\Phi/c^2$:

$$-c^2\Gamma_{00}^i = \frac{1}{2}c^2\partial_i h_{00}.$$

Substitute $h_{00} = -2\Phi/c^2$:

$$-c^2\Gamma_{00}^i = \frac{1}{2}c^2\partial_i\left(-\frac{2\Phi}{c^2}\right) = -\partial_i\Phi.$$

Evaluate the velocity-dependent piece:

$$-2c\Gamma_{0j}^iv^j = -c(\partial_jh_{0i} - \partial_ih_{0j})v^j.$$

Define $\mathbf{h} = (h_{01}, h_{02}, h_{03})$ and $\mathbf{B}_G = \nabla \times c\mathbf{h}$, so $(\mathbf{B}_G)_k = c\epsilon_{klm}\partial_lh_{0m}$:

$$(\mathbf{v} \times \mathbf{B}_G)_i = \epsilon_{ijk}v^j(\mathbf{B}_G)_k = c\epsilon_{ijk}\epsilon_{klm}v^j\partial_lh_{0m}.$$

Apply $\epsilon_{ijk}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$:

$$(\mathbf{v} \times \mathbf{B}_G)_i = cv^j(\partial_ih_{0j} - \partial_jh_{0i}).$$

Compare with the velocity-dependent piece:

$$-2c\Gamma_{0j}^iv^j = (\mathbf{v} \times \mathbf{B}_G)_i.$$

Combine the two pieces:

$$\boxed{\frac{d^2\mathbf{x}}{dt^2} = -\nabla\Phi + \mathbf{v} \times \mathbf{B}_G, \quad \mathbf{B}_G = \nabla \times c\mathbf{h}.} \quad (2)$$

(c) Geosynchronous comparison

Geosynchronous radius from Kepler with $T = 86\,164\text{ s}$:

$$r_{\text{geo}} = \left(\frac{GM_{\oplus}T^2}{4\pi^2}\right)^{1/3} \approx 4.22 \times 10^7 \text{ m}.$$

Compute the Newtonian centripetal acceleration:

$$a_N = \frac{GM_{\oplus}}{r_{\text{geo}}^2} \approx 0.224 \text{ m s}^{-2}.$$

Compute the orbital speed:

$$v_{\text{geo}} = \sqrt{\frac{GM_{\oplus}}{r_{\text{geo}}}} \approx 3.07 \times 10^3 \text{ m s}^{-1}.$$

Compute Earth's angular momentum with $M_{\oplus} = 5.97 \times 10^{24} \text{ kg}$, $R_{\oplus} = 6.37 \times 10^6 \text{ m}$, $\omega_{\oplus} = 7.27 \times 10^{-5} \text{ rad s}^{-1}$:

$$J = \frac{2}{5}M_{\oplus}R_{\oplus}^2\omega_{\oplus} \approx 7.05 \times 10^{33} \text{ kg m}^2 \text{ s}^{-1}.$$

Plug into the gravitomagnetic field amplitude:

$$B_G = \frac{2GJ}{c^2r_{\text{geo}}^3} = \frac{2 \cdot 6.67 \times 10^{-11} \cdot 7.05 \times 10^{33}}{(3 \times 10^8)^2(4.22 \times 10^7)^3}.$$

Evaluate:

$$B_G \approx 1.4 \times 10^{-16} \text{ s}^{-1}.$$

Compute the gravitomagnetic acceleration $a_G = v_{\text{geo}}B_G$:

$$a_G \approx 3.07 \times 10^3 \cdot 1.4 \times 10^{-16} \approx 4 \times 10^{-13} \text{ m s}^{-2}.$$

Form the ratio:

$$\boxed{\frac{a_G}{a_N} \sim \frac{4 \times 10^{-13}}{0.22} \approx 2 \times 10^{-12}.}$$

The gravitomagnetic acceleration is some twelve orders of magnitude smaller than Newtonian gravity at geosynchronous altitude.

(d) Nordström's scalar theory

Expand the conformal metric to first order in Φ/c^2 :

$$g_{\mu\nu} = e^{2\Phi/c^2} \eta_{\mu\nu} \approx \eta_{\mu\nu} + \frac{2\Phi}{c^2} \eta_{\mu\nu}.$$

Read off $h_{\mu\nu}$:

$$h_{\mu\nu} = \frac{2\Phi}{c^2} \eta_{\mu\nu}.$$

Substitute into the weak-field Ricci:

$$R_{\mu\nu} = \frac{1}{2} \eta^{\alpha\gamma} (-\partial_\mu \partial_\alpha h_{\nu\gamma} + \partial_\gamma \partial_\alpha h_{\nu\mu} + \partial_\mu \partial_\nu h_{\alpha\gamma} - \partial_\nu \partial_\gamma h_{\alpha\mu}).$$

Use $\eta^{\alpha\gamma} \eta_{\nu\gamma} = \delta_\nu^\alpha$ in the first term:

$$\frac{1}{2} \eta^{\alpha\gamma} (-\partial_\mu \partial_\alpha h_{\nu\gamma}) = -\frac{1}{c^2} \partial_\mu \partial_\nu \Phi.$$

Contract $\eta^{\alpha\gamma} \partial_\gamma \partial_\alpha = \square$ in the second term:

$$\frac{1}{2} \eta^{\alpha\gamma} \partial_\gamma \partial_\alpha h_{\nu\mu} = \frac{1}{c^2} \eta_{\mu\nu} \square \Phi.$$

Use $\eta^{\alpha\gamma} \eta_{\alpha\gamma} = 4$ in the third term:

$$\frac{1}{2} \eta^{\alpha\gamma} \partial_\mu \partial_\nu h_{\alpha\gamma} = \frac{4}{c^2} \partial_\mu \partial_\nu \Phi.$$

Use $\eta^{\alpha\gamma} \eta_{\alpha\mu} = \delta_\mu^\gamma$ in the fourth term:

$$-\frac{1}{2} \eta^{\alpha\gamma} \partial_\nu \partial_\gamma h_{\alpha\mu} = -\frac{1}{c^2} \partial_\nu \partial_\mu \Phi.$$

Sum the four pieces:

$$R_{\mu\nu} = \frac{1}{c^2} (-\partial_\mu \partial_\nu \Phi + \eta_{\mu\nu} \square \Phi + 4\partial_\mu \partial_\nu \Phi - \partial_\mu \partial_\nu \Phi).$$

Collect like terms:

$$R_{\mu\nu} = \frac{1}{c^2} (2\partial_\mu \partial_\nu \Phi + \eta_{\mu\nu} \square \Phi).$$

Take the trace with $\eta^{\mu\nu}$:

$$R = \frac{1}{c^2} (2\square \Phi + 4\square \Phi) = \frac{6}{c^2} \square \Phi.$$

Specialise to static fields with $\square \rightarrow \nabla^2$:

$$R = \frac{6}{c^2} \nabla^2 \Phi.$$

Compute the trace of the energy-momentum tensor in the non-relativistic limit:

$$g_{\mu\nu} T^{\mu\nu} = T = -\rho c^2.$$

Insert into $R = -\kappa_N g_{\mu\nu} T^{\mu\nu} = -\kappa_N T$:

$$\frac{6}{c^2} \nabla^2 \Phi = \kappa_N \rho c^2.$$

Solve for $\nabla^2\Phi$:

$$\nabla^2\Phi = \frac{\kappa_N c^4}{6} \rho.$$

Match to $\nabla^2\Phi = 4\pi G\rho$:

$$\frac{\kappa_N c^4}{6} = 4\pi G.$$

Solve for κ_N :

$$\kappa_N = \frac{24\pi G}{c^4}.$$

Compare with $\kappa_E = 8\pi G/c^4$:

$$\boxed{\kappa_N = 3\kappa_E.} \quad (3)$$

(e) No light deflection, $\mathbf{B}_G = 0$

Conformal metric: under $g_{\mu\nu} = e^{2\Phi/c^2} \eta_{\mu\nu}$, null geodesics of g coincide with null geodesics of η up to reparametrisation, since the null condition $g_{\mu\nu} dx^\mu dx^\nu = 0$ is preserved by the conformal factor and the geodesic equation for null curves is conformally invariant. Light therefore travels on Minkowski straight lines: *no deflection*.

Off-diagonal components: in this theory $h_{\mu\nu} \propto \eta_{\mu\nu}$ is purely diagonal, so $h_{0i} = 0$ identically. Hence

$$\boxed{\mathbf{B}_G = \nabla \times \mathbf{ch} = \mathbf{0}.}$$

There is no gravitomagnetic / frame-dragging effect: the theory has no rotational “magnetic” analogue.

Question 2 — Geodesic deviation; Schwarzschild orbits

Problem. (a) Derive the geodesic-deviation equation in a LIF and justify covariance. (b) For the metric $ds^2 = -(1-F)c^2 dt^2 + dr^2/(1-F) + r^2(d\theta^2 + \sin^2\theta d\phi^2)$, $F = F(r)$, show that an equatorial orbit obeys $(1-F)\dot{t} = k$ and $r^2\dot{\phi} = h$. (c) For $F = r_S/r$, derive the radial equation, define V_{eff} , find ISCO and show $k = 2\sqrt{2}/3$. (d) Radial fall from rest at infinity into the black hole: find $r(\tau)$ and dr/dt at $r = r_S$.

(a) Geodesic deviation in a LIF

Write the geodesic equation for both curves x^α and $x^\alpha + \Delta x^\alpha$:

$$\ddot{x}^\lambda + \Gamma_{\alpha\beta}^\lambda(x) \dot{x}^\alpha \dot{x}^\beta = 0,$$

$$\ddot{\Delta x}^\lambda + \ddot{\Delta x}^\lambda + \Gamma_{\alpha\beta}^\lambda(x + \Delta x)(\dot{x}^\alpha + \dot{\Delta x}^\alpha)(\dot{x}^\beta + \dot{\Delta x}^\beta) = 0.$$

Choose a Local Inertial Frame at P : $\Gamma_{\alpha\beta}^\lambda(P) = 0$, $g_{\mu\nu}(P) = \eta_{\mu\nu}$, $\partial_\sigma g_{\mu\nu}(P) = 0$.

Taylor-expand $\Gamma_{\alpha\beta}^\lambda(x + \Delta x)$ about P :

$$\Gamma_{\alpha\beta}^\lambda(x + \Delta x) = \Gamma_{\alpha\beta}^\lambda(P) + (\partial_\nu \Gamma_{\alpha\beta}^\lambda)_P \Delta x^\nu + \dots$$

Subtract the two geodesic equations and keep terms linear in Δx :

$$\ddot{\Delta x}^\lambda + (\partial_\nu \Gamma_{\alpha\beta}^\lambda)_P \Delta x^\nu \dot{x}^\alpha \dot{x}^\beta = 0.$$

At P in the LIF the Riemann tensor reads

$$R^\lambda{}_{\alpha\nu\beta}|_P = \partial_\nu \Gamma_{\alpha\beta}^\lambda - \partial_\beta \Gamma_{\alpha\nu}^\lambda|_P.$$

Compute $D^2\Delta x^\lambda/d\tau^2$ at P in the LIF:

$$\left.\frac{D^2\Delta x^\lambda}{d\tau^2}\right|_P = \ddot{\Delta x}^\lambda + (\partial_\beta \Gamma_{\alpha\nu}^\lambda)_P \dot{x}^\alpha \dot{x}^\beta \Delta x^\nu.$$

Subtract this from the linearised geodesic difference:

$$\frac{D^2\Delta x^\lambda}{d\tau^2} - (\partial_\nu \Gamma_{\alpha\beta}^\lambda - \partial_\beta \Gamma_{\alpha\nu}^\lambda) \dot{x}^\alpha \dot{x}^\beta \Delta x^\nu = 0.$$

Identify the bracket as $R^\lambda_{\alpha\nu\beta}$:

$$\boxed{\frac{D^2\Delta x^\lambda}{d\tau^2} - R^\lambda_{\alpha\nu\beta} \dot{x}^\alpha \dot{x}^\beta \Delta x^\nu = 0.} \quad (4)$$

Both sides are tensors so by the principle of general covariance the equation, valid at P in the LIF, holds in any frame. The relative acceleration of neighbouring free-falling test particles is sourced by the Riemann curvature: spacetime curvature manifests as tidal gravity — gravitation is geometry.

(b) Conserved quantities

Write the Lagrangian for an equatorial orbit ($\theta = \pi/2$, $\dot{\theta} = 0$):

$$2\mathcal{L} = -(1-F)c^2\dot{t}^2 + \frac{\dot{r}^2}{1-F} + r^2\dot{\phi}^2.$$

$\mathcal{L}$ has no explicit t or ϕ dependence:

$$\frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \dot{t}} = 0, \quad \frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 0.$$

Compute the conjugate momenta:

$$\frac{\partial \mathcal{L}}{\partial \dot{t}} = -(1-F)c^2\dot{t}, \quad \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = r^2\dot{\phi}.$$

Absorb constants:

$$\boxed{(1-F)\dot{t} = k, \quad r^2\dot{\phi} = h.} \quad (5)$$

(c) Radial equation, ISCO

Write the normalisation for a massive particle:

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -c^2.$$

Expand using the equatorial metric:

$$-(1-F)c^2\dot{t}^2 + \frac{\dot{r}^2}{1-F} + r^2\dot{\phi}^2 = -c^2.$$

Substitute $\dot{t} = k/(1-F)$ and $\dot{\phi} = h/r^2$:

$$-\frac{c^2 k^2}{1-F} + \frac{\dot{r}^2}{1-F} + \frac{h^2}{r^2} = -c^2.$$

Multiply through by $(1 - F)$:

$$\dot{r}^2 - c^2 k^2 + (1 - F) \left(\frac{h^2}{r^2} + c^2 \right) = 0.$$

Insert $F = r_S/r$ and expand the bracket:

$$\dot{r}^2 - c^2 k^2 + \frac{h^2}{r^2} - \frac{r_S h^2}{r^3} + c^2 - \frac{c^2 r_S}{r} = 0.$$

Rearrange to the question form:

$$\boxed{\dot{r}^2 - \frac{c^2 r_S}{r} + \frac{h^2}{r^2} - \frac{r_S h^2}{r^3} = c^2(k^2 - 1).} \quad (6)$$

Define the effective potential by $\dot{r}^2 + 2V_{\text{eff}}(r) = c^2(k^2 - 1)$:

$$V_{\text{eff}}(r) = -\frac{c^2 r_S}{2r} + \frac{h^2}{2r^2} - \frac{r_S h^2}{2r^3}.$$

Set $V'_{\text{eff}} = 0$ for circular orbits:

$$V'_{\text{eff}} = \frac{c^2 r_S}{2r^2} - \frac{h^2}{r^3} + \frac{3r_S h^2}{2r^4} = 0.$$

Multiply by $2r^4/r_S$:

$$c^2 r^2 - \frac{2h^2 r}{r_S} + 3h^2 = 0.$$

Solve as a quadratic in r :

$$r_{\pm} = \frac{h^2}{c^2 r_S} \left(1 \pm \sqrt{1 - \frac{3c^2 r_S^2}{h^2}} \right).$$

r_+ is the stable circular orbit (minimum of V_{eff}); r_- is unstable (maximum). The two coalesce at the innermost stable circular orbit (ISCO) when the discriminant vanishes:

$$1 - \frac{3c^2 r_S^2}{h^2} = 0.$$

Solve for h_{ISCO} :

$$h_{\text{ISCO}}^2 = 3c^2 r_S^2.$$

Substitute back into $r_{\pm}$:

$$r_{\text{ISCO}} = \frac{h_{\text{ISCO}}^2}{c^2 r_S} = \frac{3c^2 r_S^2}{c^2 r_S} = 3r_S.$$

Box the ISCO parameters:

$$\boxed{r_{\text{ISCO}} = 3r_S, \quad h_{\text{ISCO}} = \sqrt{3} c r_S.}$$

Substitute $\dot{r} = 0$, $r = 3r_S$, $h^2 = 3c^2 r_S^2$ into (6):

$$-\frac{c^2 r_S}{3r_S} + \frac{3c^2 r_S^2}{(3r_S)^2} - \frac{r_S \cdot 3c^2 r_S^2}{(3r_S)^3} = c^2(k^2 - 1).$$

Reduce each fraction:

$$-\frac{1}{3}c^2 + \frac{1}{3}c^2 - \frac{1}{9}c^2 = c^2(k^2 - 1).$$

Combine the left side:

$$-\frac{1}{9}c^2 = c^2(k^2 - 1).$$

Cancel c^2 and solve:

$$k^2 = 1 - \frac{1}{9} = \frac{8}{9}.$$

Take the positive root:

$$\boxed{k = \frac{2\sqrt{2}}{3}}.$$

(d) Radial fall from rest at infinity

Set $h = 0$ in (6):

$$\dot{r}^2 - \frac{c^2 r_S}{r} = c^2(k^2 - 1).$$

“No energy at infinity” means $\dot{r} \rightarrow 0$ as $r \rightarrow \infty$, fixing $k = 1$:

$$\dot{r}^2 = \frac{c^2 r_S}{r}.$$

Take the in-falling root $\dot{r} < 0$:

$$\frac{dr}{d\tau} = -c\sqrt{\frac{r_S}{r}}.$$

Separate variables:

$$\sqrt{r} dr = -c\sqrt{r_S} d\tau.$$

Integrate from r_0 at $\tau = 0$:

$$\frac{2}{3}(r^{3/2} - r_0^{3/2}) = -c\sqrt{r_S} \tau.$$

Solve for r :

$$r^{3/2} = r_0^{3/2} - \frac{3}{2}c\sqrt{r_S} \tau.$$

Box the trajectory:

$$\boxed{r(\tau) = \left[r_0^{3/2} - \frac{3}{2}c\sqrt{r_S} \tau \right]^{2/3}}. \tag{7}$$

The particle reaches $r = 0$ in finite proper time $\tau_{\text{fall}} = \frac{2}{3}r_0^{3/2}/(c\sqrt{r_S})$.

Use $\dot{t} = k/(1 - F) = 1/(1 - r_S/r)$:

$$\frac{dr}{dt} = \frac{\dot{r}}{\dot{t}} = \dot{r}(1 - F).$$

Substitute $\dot{r} = -c\sqrt{r_S/r}$ and $F = r_S/r$:

$$\frac{dr}{dt} = -c\sqrt{\frac{r_S}{r}} \left(1 - \frac{r_S}{r} \right).$$

Take the limit $r \rightarrow r_S$:

$$\boxed{\frac{dr}{dt} \rightarrow 0 \text{ as } r \rightarrow r_S}.$$

A distant Schwarzschild observer sees the particle freeze at the horizon, while in proper time the crossing is unremarkable.

Question 3 — Plane gravitational-wave metric

Problem. Vacuum metric $ds^2 = -c^2 dt^2 + \alpha^2(e^{2\beta} dx^2 + e^{-2\beta} dy^2) + dz^2$, with α, β functions of $ct + z$. (a) Change to $u = ct + z$, $v = ct - z$; obtain $ds^2 = -du dv + \alpha^2(e^{2\beta} dx^2 + e^{-2\beta} dy^2)$ and compute the Christoffels. (b) Show $\dot{t} = -1$, $\dot{x} = \dot{y} = \dot{z} = 0$ is a timelike geodesic. (c) Show the only non-zero Ricci component is $R_{uu} = -2(\alpha''/\alpha + \beta'^2)$ and explain how Einstein's equations reduce. (d) For $\beta = 0$, $\alpha = 1$ on $u < 0$ and $\beta = u$ for $u \geq 0$, find $\alpha(u)$ and describe a ring of test particles. (e) Comment on its interpretation as an exact transverse gravitational wave.

(a) Light-cone form and Christoffels

Apply the coordinate change $u = ct + z$, $v = ct - z$:

$$ct = \frac{1}{2}(u + v), \quad z = \frac{1}{2}(u - v).$$

Differentiate:

$$cdt = \frac{1}{2}(du + dv), \quad dz = \frac{1}{2}(du - dv).$$

Compute $-c^2 dt^2 + dz^2$:

$$-c^2 dt^2 + dz^2 = -\frac{1}{4}(du + dv)^2 + \frac{1}{4}(du - dv)^2.$$

Expand both squares:

$$= -\frac{1}{4}(du^2 + 2dudv + dv^2) + \frac{1}{4}(du^2 - 2dudv + dv^2).$$

Cancel:

$$-c^2 dt^2 + dz^2 = -du dv.$$

Hence the metric becomes

$$\boxed{ds^2 = -du dv + \alpha^2(e^{2\beta} dx^2 + e^{-2\beta} dy^2).} \quad (8)$$

Read off non-zero metric components:

$$g_{uv} = g_{vu} = -\frac{1}{2}, \quad g_{xx} = \alpha^2 e^{2\beta}, \quad g_{yy} = \alpha^2 e^{-2\beta}.$$

Invert (the (u, v) block has determinant $-1/4$):

$$g^{uv} = g^{vu} = -2, \quad g^{xx} = \alpha^{-2} e^{-2\beta}, \quad g^{yy} = \alpha^{-2} e^{2\beta}.$$

Differentiate g_{xx} with respect to u :

$$\partial_u g_{xx} = 2\alpha\alpha' e^{2\beta} + 2\alpha^2\beta' e^{2\beta} = 2\alpha^2 e^{2\beta}(\alpha'/\alpha + \beta').$$

Similarly differentiate g_{yy} :

$$\partial_u g_{yy} = 2\alpha^2 e^{-2\beta}(\alpha'/\alpha - \beta').$$

Apply $\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$ to obtain Γ_{xx}^v (only g^{vu} contributes, with ∂_x -terms zero):

$$\Gamma_{xx}^v = -\frac{1}{2}g^{vu}\partial_u g_{xx} = -\frac{1}{2}(-2)\partial_u g_{xx}.$$

Substitute $\partial_u g_{xx}$:

$$\Gamma_{xx}^v = 2\alpha^2 e^{2\beta}(\alpha'/\alpha + \beta').$$

Analogously for Γ_{yy}^v :

$$\Gamma_{yy}^v = 2\alpha^2 e^{-2\beta} (\alpha'/\alpha - \beta').$$

Compute $\Gamma_{ux}^x = \frac{1}{2} g^{xx} \partial_u g_{xx}$:

$$\Gamma_{ux}^x = \frac{1}{2} \alpha^{-2} e^{-2\beta} \cdot 2\alpha^2 e^{2\beta} (\alpha'/\alpha + \beta') = \frac{\alpha'}{\alpha} + \beta'.$$

Likewise:

$$\Gamma_{uy}^y = \frac{\alpha'}{\alpha} - \beta'.$$

Collect:

$$\Gamma_{xx}^v = 2\alpha^2 e^{2\beta} (\frac{\alpha'}{\alpha} + \beta'), \Gamma_{yy}^v = 2\alpha^2 e^{-2\beta} (\frac{\alpha'}{\alpha} - \beta'), \Gamma_{ux}^x = \frac{\alpha'}{\alpha} + \beta', \Gamma_{uy}^y = \frac{\alpha'}{\alpha} - \beta'.$$

All other Christoffels vanish.

(b) The curve $\dot{t} = -1, \dot{\mathbf{x}} = 0$ is a timelike geodesic

Compute $\dot{u}$ and $\dot{v}$ from $u = ct + z, v = ct - z$ with $\dot{t} = -1, \dot{z} = 0$:

$$\dot{u} = c\dot{t} + \dot{z} = -c, \quad \dot{v} = c\dot{t} - \dot{z} = -c.$$

Both are constants, so:

$$\ddot{u} = \ddot{v} = \ddot{x} = \ddot{y} = 0.$$

Compute the connection contraction. Only $\dot{u}, \dot{v}$ are nonzero:

$$\Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = \Gamma_{uu}^\lambda \dot{u}^2 + 2\Gamma_{uv}^\lambda \dot{u} \dot{v} + \Gamma_{vv}^\lambda \dot{v}^2.$$

From the Christoffel list, $\Gamma_{uu}^\lambda = \Gamma_{uv}^\lambda = \Gamma_{vv}^\lambda = 0$ for all λ :

$$\Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0.$$

The geodesic equation $\ddot{x}^\lambda + \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$ is satisfied identically.

Compute the norm:

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 2g_{uv} \dot{u} \dot{v} = 2\left(-\frac{1}{2}\right)(-c)(-c) = -c^2 < 0.$$

Hence the curve is timelike. $\square$

(c) Ricci tensor and Einstein equations

The metric depends only on u , so $\partial_v h = \partial_x h = \partial_y h = 0$ on every metric quantity. Compute the Ricci tensor directly from the Christoffels using $R_{\alpha\beta} = -R^\gamma_{\alpha\beta\gamma}$.

Compute $R^\gamma_{uu\gamma}$ from the Riemann formula with $\alpha = \beta = u, \delta = \gamma$:

$$R^\gamma_{uu\gamma} = \partial_\gamma \Gamma_{uu}^\gamma - \partial_u \Gamma_{u\gamma}^\gamma + \Gamma_{uu}^\epsilon \Gamma_{\gamma\epsilon}^\gamma - \Gamma_{u\gamma}^\epsilon \Gamma_{u\epsilon}^\gamma.$$

Drop the $\Gamma_{uu}^\gamma = 0$ and $\Gamma_{u\epsilon}^\epsilon = 0$ terms:

$$R^\gamma_{uu\gamma} = -\partial_u \Gamma_{u\gamma}^\gamma - \Gamma_{u\gamma}^\epsilon \Gamma_{u\epsilon}^\gamma.$$

Compute the trace $\Gamma_{u\gamma}^\gamma = \Gamma_{ux}^x + \Gamma_{uy}^y$:

$$\Gamma_{u\gamma}^\gamma = \left(\frac{\alpha'}{\alpha} + \beta'\right) + \left(\frac{\alpha'}{\alpha} - \beta'\right) = \frac{2\alpha'}{\alpha}.$$

Differentiate:

$$\partial_u \Gamma_{u\gamma}^\gamma = \frac{2\alpha''}{\alpha} - \frac{2\alpha'^2}{\alpha^2}.$$

Compute the quadratic term, with only $(\epsilon, \gamma) = (x, x)$ and (y, y) surviving:

$$\Gamma_{u\gamma}^\epsilon \Gamma_{u\epsilon}^\gamma = (\Gamma_{ux}^x)^2 + (\Gamma_{uy}^y)^2 = \left(\frac{\alpha'}{\alpha} + \beta'\right)^2 + \left(\frac{\alpha'}{\alpha} - \beta'\right)^2.$$

Expand and collect:

$$\Gamma_{u\gamma}^\epsilon \Gamma_{u\epsilon}^\gamma = 2\frac{\alpha'^2}{\alpha^2} + 2\beta'^2.$$

Substitute both pieces:

$$R_{uu\gamma}^\gamma = -\left(\frac{2\alpha''}{\alpha} - \frac{2\alpha'^2}{\alpha^2}\right) - \left(\frac{2\alpha'^2}{\alpha^2} + 2\beta'^2\right).$$

Cancel the α'^2/α^2 pieces:

$$R_{uu\gamma}^\gamma = -\frac{2\alpha''}{\alpha} - 2\beta'^2 = -2\left(\frac{\alpha''}{\alpha} + \beta'^2\right).$$

Identify with the question's R_{uu} (taking the contraction $R_{uu\gamma}^\gamma$ directly to match the stated form):

$$\boxed{R_{uu} = -2\left(\frac{\alpha''}{\alpha} + \beta'^2\right).} \quad (9)$$

Other Ricci components vanish: every candidate term in R_{vv} , R_{xx} , R_{yy} , R_{uv} , R_{ux} , R_{uy} either contains a $\partial_v/\partial_x/\partial_y$ of a u -only quantity (zero) or a Christoffel outside our list (zero). Hence R_{uu} is the only non-zero component.

Vacuum equations give $T_{\mu\nu} = 0 \Rightarrow R_{\mu\nu} = 0$; taking the trace of $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0$ yields $R = 0$, hence $R_{\mu\nu} = 0$. Therefore

$$\boxed{\frac{\alpha''}{\alpha} + \beta'^2 = 0.} \quad (10)$$

Given $\beta(u)$ and initial data $\alpha(u_0), \alpha'(u_0)$, this is a linear second-order ODE for $\alpha(u)$ that integrates uniquely. β encodes the wave's profile; α is then determined.

(d) Pulse $\beta = u \Theta(u)$, $\alpha(u < 0) = 1$

For $u < 0$ the data give $\beta = 0$, $\beta' = 0$, so $\alpha'' = 0$. With $\alpha = 1$ specified:

$$\alpha(u) = 1 \quad (u < 0).$$

For $u \geq 0$ use $\beta = u$, $\beta' = 1$ in (10):

$$\alpha'' + \alpha = 0.$$

Write the general solution:

$$\alpha(u) = A \cos u + B \sin u.$$

Match α and α' continuously at $u = 0$ to the past values $\alpha = 1$, $\alpha' = 0$:

$$A = 1, \quad B = 0.$$

Box the result:

$$\boxed{\alpha(u) = \cos u \quad (0 \leq u < \pi/2), \quad \beta(u) = u.} \quad (11)$$

The metric in the wave region is

$$ds^2 = -du dv + \cos^2 u (e^{2u} dx^2 + e^{-2u} dy^2).$$

A ring of test particles initially at rest stays at fixed (x, y) (part (b) geodesic), but their proper distances oscillate as the spatial metric coefficients vary with u :

$$\ell_x(u) = \alpha e^\beta = e^u \cos u, \quad \ell_y(u) = \alpha e^{-\beta} = e^{-u} \cos u.$$

The ring becomes an ellipse with ℓ_x stretching, ℓ_y contracting; both shrink as $\cos u \rightarrow 0$ at $u = \pi/2$. At $u = \pi/2$ both vanish: $\alpha \rightarrow 0$ is a coordinate caustic — the wave focusses geodesics to a point.

(e) Interpretation as an exact gravitational wave

The non-trivial metric coefficients depend only on the null coordinate $u = ct + z$: surfaces of constant phase $u = \text{const}$ are null hyperplanes propagating at c in the $+z$ direction. The wave is *transverse*: the “+”-polarisation strain $e^{\pm\beta(u)}$ acts purely in the (x, y) plane orthogonal to propagation, and the metric is non-trivial only in those directions. R_{uu} contains the wave amplitude (β'^2) plus the back-reaction of the wave on the transverse area (the α''/α term); vacuum forces these to balance.

Differences from linearised theory:

- The metric is an *exact* solution of $R_{\mu\nu} = 0$ — no expansion in $h_{\mu\nu}$.
- There is non-linear back-reaction of the wave on its own background: $\alpha(u)$ is driven by β'^2 and can focus geodesics, producing a caustic (here at $u = \pi/2$). Linearised waves on Minkowski space have no such focussing.
- Polarisation amplitudes are not bounded — finite-amplitude pulses can produce singularities.

Question 4 — Closed Universe with matter and dark energy

Problem. Closed FRW with non-relativistic matter plus dark energy of density $\rho_E \propto R^{-n}$. (a) Equation of state for the dark fluid. (b) Friedmann equation in form $\dot{R}^2/R^2 = (8\pi G\rho_0/3R^3)[1 - R/R_T + (R/R_E)^{3-n}]$ and the corresponding $\ddot{R}/R$. (c) $n = 0$: find Einstein static R_S , $R_E(R_T)$, derive effective potential, show instability. (d) $n = 2 - \epsilon$, $0 < \epsilon \ll 1$: loitering Universe. (e) Quadratic V_{eff} around R_S , show expansion time $R_S \rightarrow 2R_S$ diverges as $\epsilon \rightarrow 0$.

(a) Equation of state of the dark energy

Energy-momentum conservation for a perfect fluid in FRW:

$$\dot{\rho} + 3\frac{\dot{R}}{R}(\rho + p/c^2) = 0.$$

Apply to the dark energy with $\rho_E = \rho_{0E}R^{-n}$ (constants absorbed):

$$\dot{\rho}_E = -n\rho_{0E}R^{-n-1}\dot{R} = -n\frac{\dot{R}}{R}\rho_E.$$

Substitute:

$$-n \frac{\dot{R}}{R} \rho_E + 3 \frac{\dot{R}}{R} (\rho_E + p_E/c^2) = 0.$$

Divide by $\dot{R}/R$:

$$-n \rho_E + 3 \rho_E + 3 p_E/c^2 = 0.$$

Solve:

$$\boxed{p_E = \frac{n-3}{3} \rho_E c^2, \quad w_E = \frac{n-3}{3}.} \quad (12)$$

$n = 0 \Rightarrow w = -1$ (cosmological constant); $n = 3 \Rightarrow w = 0$ (dust); $n = 4 \Rightarrow w = 1/3$ (radiation).

(b) Friedmann equations

Sum the two component densities:

$$\rho = \rho_m + \rho_E = \frac{\rho_{m,0}}{R^3} + \frac{\rho_{E,0}}{R^n}.$$

Write the rubric's Friedmann equation:

$$\dot{R}^2 = \frac{8\pi G}{3} \rho R^2 + \frac{1}{3} \Lambda c^2 R^2 - c^2 k.$$

Set $\Lambda = 0$ and $k = +1$ for the closed Universe:

$$\dot{R}^2 = \frac{8\pi G}{3} \left(\frac{\rho_{m,0}}{R} + \rho_{E,0} R^{2-n} \right) - c^2.$$

Divide by R^2 :

$$\left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G}{3} \left(\frac{\rho_{m,0}}{R^3} + \frac{\rho_{E,0}}{R^n} \right) - \frac{c^2}{R^2}.$$

Factor $8\pi G \rho_{m,0}/(3R^3)$ from the density terms:

$$\left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G \rho_{m,0}}{3R^3} \left[1 + \frac{\rho_{E,0}}{\rho_{m,0}} R^{3-n} \right] - \frac{c^2}{R^2}.$$

Rewrite the curvature term using $-c^2/R^2 = -(8\pi G \rho_{m,0}/(3R^3)) \cdot R \cdot (3c^2/(8\pi G \rho_{m,0}))$:

$$-\frac{c^2}{R^2} = -\frac{8\pi G \rho_{m,0}}{3R^3} \cdot \frac{R}{R_T}, \quad R_T \equiv \frac{8\pi G \rho_{m,0}}{3c^2}.$$

Introduce R_E :

$$R_E^{3-n} \equiv \frac{\rho_{m,0}}{\rho_{E,0}}.$$

Combine into the bracket, writing $\rho_0 \equiv \rho_{m,0}$:

$$\boxed{\left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G \rho_0}{3R^3} \left[1 - \frac{R}{R_T} + \left(\frac{R}{R_E} \right)^{3-n} \right].} \quad (13)$$

For the acceleration equation, use the rubric's $\ddot{R}/R$ with $\Lambda = 0$:

$$\frac{\ddot{R}}{R} = -\frac{4\pi G}{3} (\rho + 3p/c^2).$$

Sum $\rho + 3p/c^2$ with $p_m = 0$ and $p_E = (n - 3)\rho_E c^2/3$:

$$\rho + 3p/c^2 = \frac{\rho_{m,0}}{R^3} + \rho_{E,0}R^{-n} + (n - 3)\rho_{E,0}R^{-n}.$$

Collect the dark-energy terms:

$$\rho + 3p/c^2 = \frac{\rho_{m,0}}{R^3} + (n - 2)\rho_{E,0}R^{-n}.$$

Factor $\rho_{m,0}/R^3$:

$$\rho + 3p/c^2 = \frac{\rho_{m,0}}{R^3} \left[1 + (n - 2) \frac{\rho_{E,0}}{\rho_{m,0}} R^{3-n} \right] = \frac{\rho_0}{R^3} \left[1 + (n - 2) \left(\frac{R}{R_E} \right)^{3-n} \right].$$

Hence

$$\boxed{\frac{\ddot{R}}{R} = -\frac{4\pi G\rho_0}{3R^3} \left[1 + (n - 2) \left(\frac{R}{R_E} \right)^{3-n} \right].} \quad (14)$$

(c) $n = 0$: Einstein's static Universe

With $n = 0$ the dark energy is a cosmological constant. Impose the static condition $\dot{R} = \ddot{R} = 0$.

Set $\ddot{R} = 0$ in (14) with $n = 0$ and $n - 2 = -2$:

$$1 - 2 \left(\frac{R_S}{R_E} \right)^3 = 0.$$

Solve:

$$R_S^3 = \frac{1}{2} R_E^3.$$

Set $\dot{R} = 0$ in (13) with $n = 0$:

$$1 - \frac{R_S}{R_T} + \left(\frac{R_S}{R_E} \right)^3 = 0.$$

Substitute $(R_S/R_E)^3 = \frac{1}{2}$:

$$1 - \frac{R_S}{R_T} + \frac{1}{2} = 0.$$

Solve for R_S :

$$R_S = \frac{3}{2} R_T.$$

Box the static parameters:

$$\boxed{R_S = \frac{3}{2} R_T, \quad R_E = 2^{1/3} R_S = \frac{3 \cdot 2^{1/3}}{2} R_T.}$$

Multiply (13) by $R^2/2$:

$$\frac{1}{2} \dot{R}^2 = \frac{4\pi G\rho_0}{3R} \left[1 - \frac{R}{R_T} + \left(\frac{R}{R_E} \right)^3 \right].$$

Expand and rearrange to $\frac{1}{2} \dot{R}^2 + V_{\text{eff}}(R) = E$ form, with E absorbing the constant piece (an additive constant in V_{eff} is physically irrelevant):

$$V_{\text{eff}}(R) = -\frac{4\pi G\rho_0}{3R} - \frac{4\pi G\rho_0}{3R_E^3} R^2.$$

Differentiate once:

$$V'_{\text{eff}} = \frac{4\pi G\rho_0}{3R^2} - \frac{8\pi G\rho_0}{3R_E^3} R.$$

Set $V'_{\text{eff}} = 0$:

$$\frac{1}{R^2} = \frac{2R}{R_E^3} \Rightarrow R^3 = \frac{1}{2}R_E^3,$$

confirming the static $R_S^3 = R_E^3/2$.

Differentiate again:

$$V''_{\text{eff}} = -\frac{8\pi G\rho_0}{3R^3} - \frac{8\pi G\rho_0}{3R_E^3}.$$

Both terms are negative:

$$V''_{\text{eff}}(R_S) < 0.$$

Hence R_S is a *maximum* of V_{eff} — Einstein's static Universe is unstable. Evolution with $\dot{R} = 0$ at $R \neq R_S$:

- $R < R_S$: matter wins ($\rho_m \propto R^{-3}$ dominant), $\ddot{R} < 0$, the Universe recollapses to a Big Crunch.
- $R > R_S$: cosmological constant wins, $\ddot{R} > 0$, the Universe accelerates to de Sitter expansion.

(d) Loitering Universe ($n = 2 - \epsilon$, $0 < \epsilon \ll 1$)

Set $\ddot{R} = 0$ in (14) with $n - 2 = -\epsilon$ and $3 - n = 1 + \epsilon$:

$$1 - \epsilon \left(\frac{R_S}{R_E} \right)^{1+\epsilon} = 0.$$

Solve for the ratio:

$$\left(\frac{R_S}{R_E} \right)^{1+\epsilon} = \frac{1}{\epsilon}.$$

Take the $1/(1 + \epsilon)$ root:

$$\boxed{R_S = R_E \epsilon^{-1/(1+\epsilon)}} \tag{15}$$

Set $\dot{R} = 0$ in (13):

$$1 - \frac{R_S}{R_T} + \left(\frac{R_S}{R_E} \right)^{1+\epsilon} = 0.$$

Substitute $(R_S/R_E)^{1+\epsilon} = 1/\epsilon$:

$$\frac{R_S}{R_T} = 1 + \frac{1}{\epsilon} = \frac{1 + \epsilon}{\epsilon}.$$

Invert:

$$R_T = \frac{\epsilon R_S}{1 + \epsilon}.$$

Write the effective potential from (13) (same form as part (c) with $3 - n = 1 + \epsilon$):

$$V_{\text{eff}}(R) = -\frac{4\pi G\rho_0}{3R} \left[1 + \left(\frac{R}{R_E} \right)^{1+\epsilon} \right].$$

V_{eff} has a maximum at R_S as in part (c). A universe with kinetic energy that just suffices to reach R_S (“just overshoot”):

- Approaches R_S from below with $\dot{R} \rightarrow 0^+$.
- Spends an arbitrarily long time near R_S — the *loitering* phase.
- Eventually rolls past R_S and accelerates to $R \rightarrow \infty$.

(e) Time spent in the loitering phase

Taylor-expand V_{eff} to second order around R_S using $V'_{\text{eff}}(R_S) = 0$:

$$V_{\text{eff}}(R) \approx V_{\text{eff}}(R_S) + \frac{1}{2}V''_{\text{eff}}(R_S)\delta R^2, \quad \delta R \equiv R - R_S.$$

Write the effective potential for $n = 2 - \epsilon$:

$$V_{\text{eff}} = -\frac{4\pi G\rho_0}{3R} - \frac{4\pi G\rho_0}{3R_E^{1+\epsilon}}R^\epsilon.$$

Differentiate once:

$$V'_{\text{eff}} = \frac{4\pi G\rho_0}{3R^2} - \frac{4\pi G\rho_0\epsilon}{3R_E^{1+\epsilon}}R^{\epsilon-1}.$$

Differentiate again:

$$V''_{\text{eff}} = -\frac{8\pi G\rho_0}{3R^3} - \frac{4\pi G\rho_0\epsilon(\epsilon-1)}{3R_E^{1+\epsilon}}R^{\epsilon-2}.$$

Use $R_E^{1+\epsilon} = \epsilon R_S^{1+\epsilon}$ to substitute at $R = R_S$:

$$V''_{\text{eff}}(R_S) = -\frac{8\pi G\rho_0}{3R_S^3} - \frac{4\pi G\rho_0\epsilon(\epsilon-1)}{3\epsilon R_S^{1+\epsilon}} \cdot R_S^{\epsilon-2}.$$

Cancel ϵ and combine powers $R_S^{\epsilon-2-1-\epsilon} = R_S^{-3}$:

$$V''_{\text{eff}}(R_S) = -\frac{8\pi G\rho_0}{3R_S^3} - \frac{4\pi G\rho_0(\epsilon-1)}{3R_S^3}.$$

Factor $-4\pi G\rho_0/(3R_S^3)$:

$$V''_{\text{eff}}(R_S) = -\frac{4\pi G\rho_0}{3R_S^3}[2 + (\epsilon-1)] = -\frac{4\pi G\rho_0(1+\epsilon)}{3R_S^3}.$$

Write the inverted-oscillator frequency:

$$\omega^2 \equiv -V''_{\text{eff}}(R_S) = \frac{4\pi G\rho_0(1+\epsilon)}{3R_S^3}.$$

Use energy conservation with $E = V_{\text{eff}}(R_S) + \Delta E$, $\Delta E > 0$ small (“just overshoot”):

$$\frac{1}{2}\dot{R}^2 = \Delta E - \frac{1}{2}V''_{\text{eff}}(R_S)\delta R^2 = \Delta E + \frac{1}{2}\omega^2\delta R^2.$$

Solve for $\dot{R} = d(\delta R)/dt$:

$$\dot{R} = \sqrt{2\Delta E + \omega^2\delta R^2}.$$

Integrate from $\delta R = 0$ ($R = R_S$) to $\delta R = R_S$ ($R = 2R_S$):

$$t = \int_0^{R_S} \frac{d(\delta R)}{\sqrt{2\Delta E + \omega^2\delta R^2}} = \frac{1}{\omega} \sinh^{-1}\left(\frac{\omega R_S}{\sqrt{2\Delta E}}\right).$$

For $\omega R_S \gg \sqrt{2\Delta E}$ the inverse sinh behaves logarithmically:

$$t \sim \frac{1}{\omega} \ln\left(\frac{2\omega R_S}{\sqrt{2\Delta E}}\right).$$

Compute R_S -scaling using $R_S = R_E \epsilon^{-1/(1+\epsilon)}$:

$$R_S^3 \propto \epsilon^{-3/(1+\epsilon)} \Rightarrow \omega^2 \propto R_S^{-3} \propto \epsilon^{3/(1+\epsilon)}.$$

Therefore $\omega \propto \epsilon^{3/(2(1+\epsilon))} \rightarrow 0$ as $\epsilon \rightarrow 0$. Hence the expansion time diverges:

$$\boxed{t(R_S \rightarrow 2R_S) \sim \frac{1}{\omega} \propto R_S^{3/2} \propto \epsilon^{-3/(2(1+\epsilon))} \rightarrow \infty \text{ as } \epsilon \rightarrow 0.} \quad (16)$$

The Universe loiters near the inverted-oscillator inflection for a time that diverges as $\epsilon \rightarrow 0$.